

Computational Chemistry Comparison and Benchmark DataBase Release 22 (May 2022) Standard
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Experimental data for CO₂ (Carbon dioxide)

22 02 02 11 45

Other names			
Anhydride carbonique; Carbon dioxide; Carbon oxide (CO ₂); Carbonic acid, gas; Carbonic anhydride; Carbonice; Dry ice; Kohlendioxyd; Kohlensaure;			
INChI	INChIKey	SMILES	IUPAC name
InChI=1S/CO2/c2-1-3	CURLTUGMZLYLDI-UHFFFAOYSA-N	O=C=O	Carbon dioxide
State	Conformation		
¹ Σ _g	D*H		

Enthalpy of formation (H_{fg}), Entropy, Integrated heat capacity (0 K to 298.15 K) (HH), Heat Capacity (C_p)

Property	Value	Uncertainty	units	Reference	Comment
H _{fg} (298.15K) Δ _f H°	-393.51	0.13	kJ mol ⁻¹	CODATA	
H _{fg} (0K) Δ _f H°	-393.14	0.13	kJ mol ⁻¹	CODATA	
Entropy (298.15K) S°	213.79	0.01	J K ⁻¹ mol ⁻¹	CODATA	
Integrated Heat Capacity (0 to 298.15K) H°-H°(T)	9.36	0.00	kJ mol ⁻¹	CODATA	
Heat Capacity (298.15K) C _p °	37.13		J K ⁻¹ mol ⁻¹	Gurvich	

Information can also be found for this species in the [NIST Chemistry Webbook](#)

Vibrational levels (cm⁻¹) ↕

Mode Number	Symmetry	Frequency			Intensity			Comment	Description
		Fundamental(cm ⁻¹)	Harmonic(cm ⁻¹)	Reference	(km mol ⁻¹)	unc.	Reference		
1	Σ _g	1333	1351	1945Herzberg				unperturbed value	CO ₂ s-stretch
2	Σ _u	2349	2396	1945Herzberg	665.0		1969Dav/Orv:163		CO ₂ a-stretch
3	Π _u	667	672	1945Herzberg	54.2		1982vibint		bend

vibrational zero-point energy: 2508.0 cm⁻¹ (from fundamental vibrations)

[Calculated vibrational frequencies](#) for CO₂ (Carbon dioxide).

Gas-phase IR spectra can be found in the NIST Chemistry Webbook [here](#).

Rotational Constants (cm⁻¹) ↕

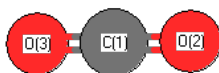
See section [I.F.4 to change rotational constant units](#)

A	B	C	reference	comment
	0.39021		1966Herzberg	

[Calculated rotational constants](#) for CO₂ (Carbon dioxide).

Product of moments of inertia I _A I _B I _C			
43.20143	amu Å ²	7.173879E-39	gm cm ²

Geometric Data

Point Group $D_{\infty h}$

Internal coordinates

distances (r) in Å, angles (a) in degrees, dihedrals (d) in degrees

Description	Value	unc.	Connectivity				Reference	Comment
			Atom 1	Atom 2	Atom 3	Atom 4		
rCO	1.162		1	2			1966Herzberg	
aOCO	180		2	1	3		1966Herzberg	

Cartesians

Atom	x (Å)	y (Å)	z (Å)
C1	0.0000	0.0000	0.0000
O2	0.0000	0.0000	1.1621
O3	0.0000	0.0000	-1.1621

Atom - Atom Distances

Distances in Å

	C1	O2	O3
C1		1.1621	1.1621
O2	1.1621		2.3242
O3	1.1621	2.3242	

[Calculated geometries](#) for CO₂ (Carbon dioxide).

Experimental Bond Angles (degrees) from cartesians

atom1	atom2	atom3	angle
O2	C1	O3	180.000

Bond descriptions

Examples: C-C single bond, C=C, double bond, C#C triple bond, C:C aromatic bond

Bond Type	Count
C=O	2

Connectivity

Atom 1	Atom 2
C1	O2
C1	O3

Electronic energy levels (cm⁻¹)

Energy (cm ⁻¹)	Degeneracy	reference	description
0	1		$1\Sigma_g$
46000	1	1966Herzberg	

Ionization Energies (eV)

Ionization Energy	I.E. unc.	vertical I.E.	v.I.E. unc.	reference
13.777	0.001	13.777		webbook

Electron Affinity (eV)

Electron Affinity	unc.	reference
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-0.600	webbook
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Proton Affinity (kJ mol⁻¹)

Proton Affinity	unc.	Product	reference	comment
540.5		HOCO ⁺	webbook	

Dipole, Quadrupole and Polarizability

Electric dipole moment $\leftrightarrow$

State	Config	State description	Conf description	Exp. min.	Dipole (Debye)				Reference	comment	Point Group	Components	
					x	y	z	total				dipole	quadrupole
1	1	$1\Sigma_g$	D _{∞h}	True				0.000	NSRDS-NBS10		D _{∞h}	0	1

Experimental dipole measurement abbreviations: MW microwave; DT Dielectric with Temperature variation; DR Indirect (usually an upper limit); MB Molecular beam

[Calculated electric dipole moments](#) for CO₂ (Carbon dioxide).

Electric quadrupole moment $\begin{smallmatrix} + & - \\ - & + \end{smallmatrix}$

State	Config	State description	Conf description	Exp. min.	Quadrupole (D Å)			Reference	comment	Point Group	Components	
					xx	yy	zz				dipole	quadrupole
1	1	$1\Sigma_g$	D _{∞h}	True	2.139	2.139	-4.278	1998Gra/Imr:49	-14.27± 0.61 E-40 C m ²	D _{∞h}	0	1

[Calculated electric quadrupole moments](#) for CO₂ (Carbon dioxide).

Electric dipole polarizability (Å³) $\leftrightarrow$

alpha	unc.	Reference
2.507		1997Oln/Can:59

[Calculated electric dipole polarizability](#) for CO₂ (Carbon dioxide).

References

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squib	reference	DOI
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1966Herzberg	Herzberg, G., Electronic spectra and electronic structure of polyatomic molecules,Van Nostrand,New York, 1966	
1969Dav/Orv:163	PR Davies, WJ Orville-Thomas "Infrared Band Intensities and Bond polarities Part1. Bond Moment constants in CO2, OCS, CS2, CSe2, and SCSe" J. Mol. Struct. 4 (1969) 163	10.1016/0022-2860(69)80052-9
1982vibint	"Vibrational Intensities in Infrared and Raman Spectroscopy" WB Person, G Zerbi, ed. Elsevier, Amsterdam, 1982	
1997Oln/Can:59	TN Olney, NM Cann, G Cooper, CE Brion, Absolute scale determination for photoabsorption spectra and the calculation of molecular properties using dipole sum-rules, Chem. Phys. 223 (1997) 59-98	10.1016/S0301-0104(97)00145-6
1998Gra/Imr:49	C Graham, DA Imrie, RE Raab "Measurement of the electric quadrupole moments of CO2, CO, N2, Cl2 and BF3" Mol. Phys. 93(1), 1998, 49-56	10.1080/002689798169429
CODATA	Cox, J.D.; Wagman, D.D.; Medvedev, V.A.CODATA Key Values for Thermodynamics. Hemisphere, New York, 1989	10.1002/bbpc.19900940121
Gurvich	Gurvich, L.V.; Veyts, I. V.; Alcock, C. B., Thermodynamic Properties of Individual Substances, Fouth Edition, Hemisphere Pub. Co., New York, 1989	
NSRDS-NBS10	R. D. Nelson Jr., D. R. Lide, A. A. Maryott "Selected Values of electric dipole moments for molecules in the gas phase" NSRDS-NBS10, 1967	10.6028/NBS.NSRDS.10
webbook	NIST Chemistry Webbook (http://webbook.nist.gov/chemistry)	10.18434/T4D303

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PPO (Phosphorus oxide phosphide)	CO2- (Carbon dioxide anion)